

# DS358 Deep Speech Processing

S. R. Mahadeva Prasanna

Professor, Dept. of DS & AI, IIIT Dharwad  
Director, IIIT Dharwad

*prasanna@iiitdwd.ac.in*

December 31, 2024



# Some Resources for Deep Speech Processing

- L.R. Rabiner and R.W. Schafer, Digital Processing of Speech Signals Pearson Education, Delhi, India, 2004
- L. R. Rabiner, B. H. Juang and B. Yegnanarayana, "Fundamentals of speech recognition", Pearson Education, 2009.
- Linus Ericsson, et al., Self-Supervised Representation Learning: Introduction, Advances and Challenges, arXiv:2110.09327
- R. Prabhavalkar, et.al., "End-to-End Speech Recognition: A Survey," in IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 32, pp. 325-351, 2024,
- Xie, Tianxin, et al. "Towards Controllable Speech Synthesis in the Era of Large Language Models: A Survey." (2024).
- Wang, S., et.al., "Overview of speaker modeling and its applications: From the lens of deep speaker representation learning" IEEE/ACM Transactions on Audio, Speech, and Language Processing.



- **Overview of Speech Processing:** Speech production, perception and comprehension, spoken language processing, human speech processing tasks, nature of speech signal, signal processing of speech, linguistics and phonetics, feature extraction. (6 hrs)
- **Machine Learning of Speech Processing:** clustering of speech features, classification tasks in speech, Gaussian mixture model (GMM), Hidden Markov Model (HMM), Support Vector Machine (SVM) and Neural Networks (NN) (6 hrs)
- **Deep Learning of Speech Processing:** Self-Supervised Learning for Speech Representation. Automatic Speech Recognition (ASR). Text to Speech (TTS) synthesis. Speaker and Language Recognition. (Details in Annexure - DSP) (30 hrs)

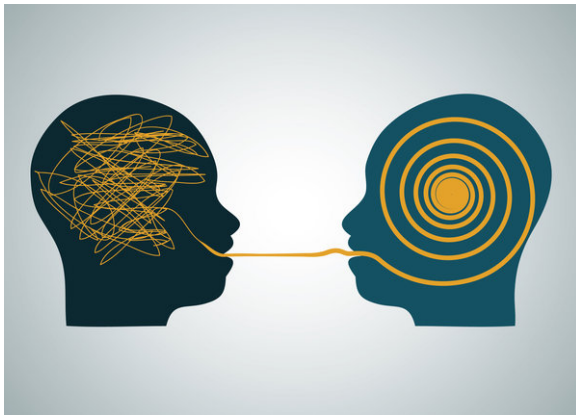
# Organization

- Human speech processing
- Natural Language : Spoken vs written
- Information sources in speech
- Human speech processing tasks
- Acoustic form of speech signal
- Electrical form of speech signal
- Digital speech

# Human Speech Processing

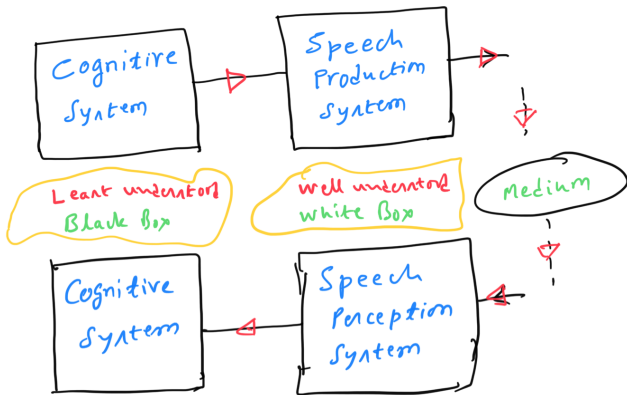
- Cognitive process
- Physiological process
- Production of speech
- Perception of speech
- Comprehension of speech

# Cognitive Processing



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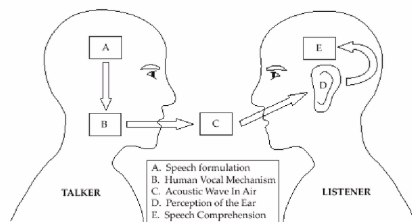
# Cognitive and Physiological Processes



# Human - Human Communication

|                 | <b>Verbal Communication</b>        | <b>Nonverbal Communication</b>     |
|-----------------|------------------------------------|------------------------------------|
| <b>Oral</b>     | Spoken Language                    | Laughing, Crying, Coughing, Etc... |
| <b>Non Oral</b> | Written Language/<br>Sign Language | Gestures, Body Language, Etc...    |

(a) Verbal vs Non-Verbal Communication<sup>1</sup>



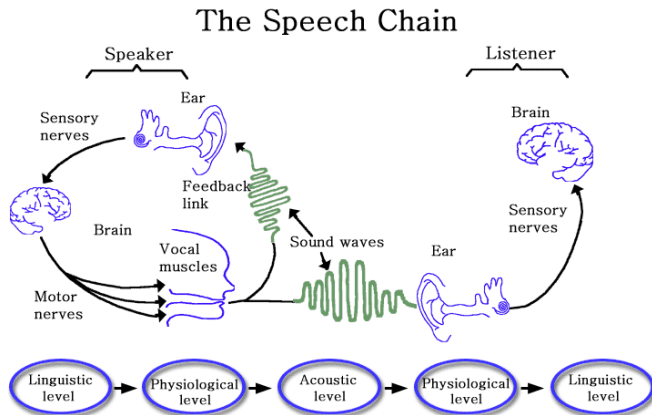
(b) Speech production, transmission, perception, comprehension<sup>2</sup>

1. <https://mytext.cnm.edu/lesson/5-1-0-defining-verbal-communication/>

2. <https://towardsdatascience.com/speech-recognition-is-hard-part-1-258e813b6eb7>

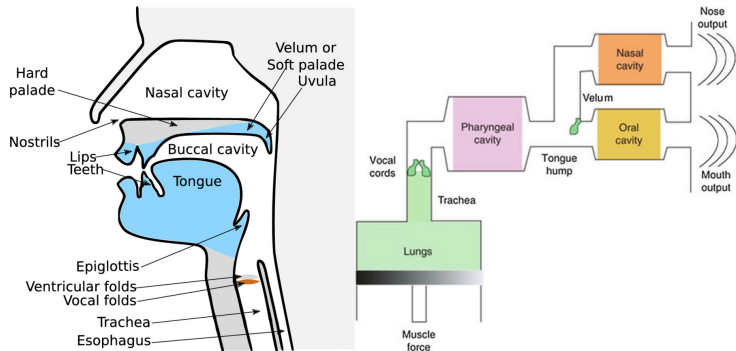


# Human Speech Communication Chain



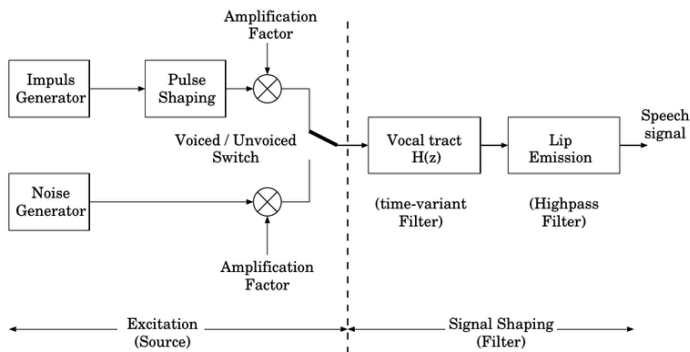
<http://indra-bohara.blogspot.com/2010/10/brief-critical-review-of-speech-chain.html>

# Feature Extraction for Acoustic Modeling



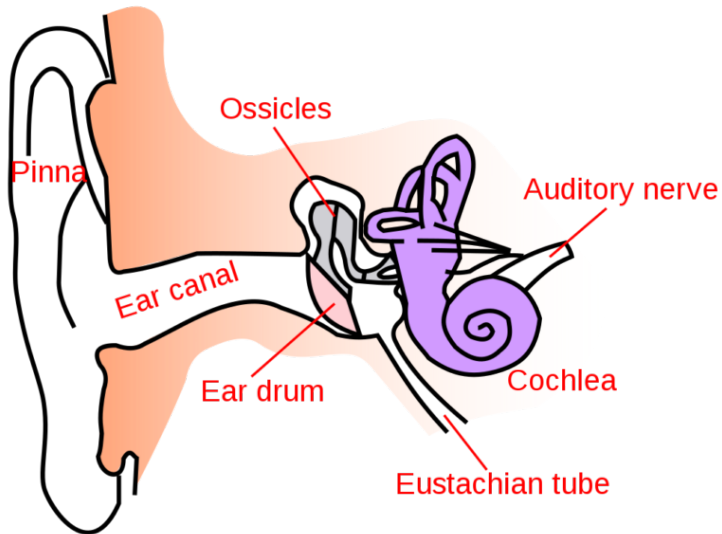
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# Feature Extraction for Acoustic Modeling



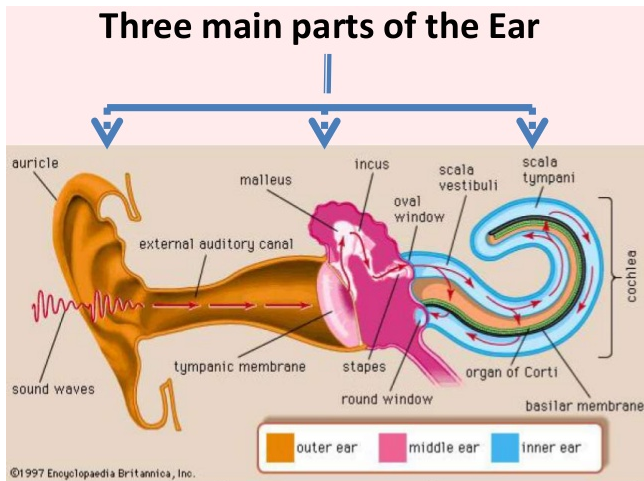
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# Speech Perception Process: Ear



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# Speech Perception Process: Acoustic to BEP



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# Natural Language

- Unless specified language refers to **natural language** used by humans
- Not to be confused with **artificial languages** used by computers
- A natural language has developed and evolved naturally, through use by human beings
- As opposed to an invented or constructed language, as a computer programming language

# Classification of Natural Language

- **Language** : Fundamental method of human communication, consisting of words used in a structured, and conventional way and conveyed by speech, writing, or gesture.
- **Spoken Language** : Human communication via speech mode
- **Written Language** : Human communication via written mode
- **Sign Language** : Human communication via gesture mode

# Components of Natural Language

- **Components of a language** : Basic units for representation, words, rules for forming phrases and sentences
- Meaning of words
- Meaning of phrases
- Meaning of sentences
- Hidden information like emotions, sentiments ...



# Components of Spoken Language

- **Phonemes or phones** : Basic units of speech
- **Syllable** : Bigger basic units of speech, combination of one or more phonemes
- Spoken words, phrases and sentences
- **Prosody** : Elements of speech that are properties of syllables and larger units of speech, including linguistic functions such as intonation, stress, and rhythm

# Hindi Sound Units

- Vowels & consonants
- Short vowels, long vowels and diphthongs
- Stop consonants, nasals, semivowels, fricatives, affricates
- Compound characters

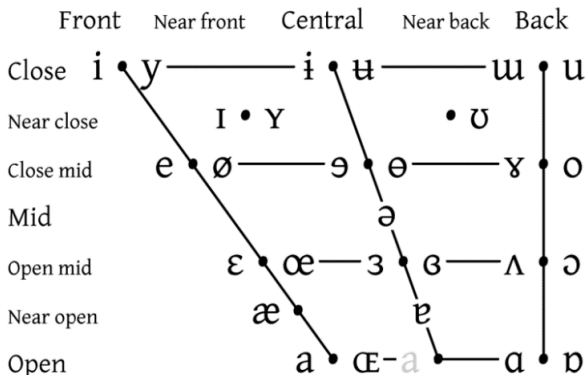
# Hindi Vowels



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# Production of Vowels

## VOWELS



Vowels at right & left of bullets are rounded & unrounded.

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# Hindi Consonants

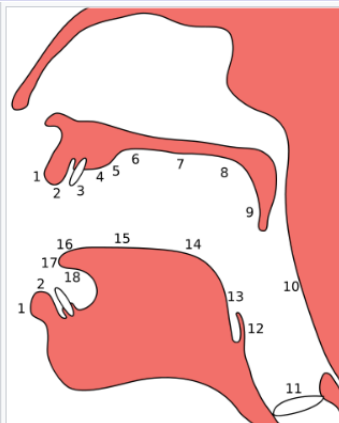


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# Different Aspects during Production

- Place of articulation (PoA)
- Manner of articulation (MoA)
- Type of vowel
- PoA : Glottal, velar, alveolar, palatal, dental, labial
- MoA : Unvoiced unaspirated (UVUA), Unvoiced aspirated (UVA), Voiced unaspirated (VUA) and Voiced aspirated (VA)
- Vowel : Front, back and middle

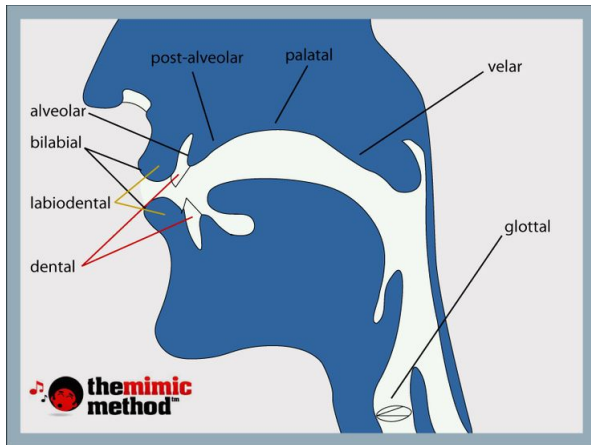
# Place of Articulation



Places of articulation (passive & active): 

1. Exo-labial, 2. Endo-labial, 3. Dental, 4. Alveolar, 5. Post-alveolar, 6. Pre-palatal, 7. Palatal, 8. Velar, 9. Uvular, 10. Pharyngeal, 11. Glottal, 12. Epiglottal, 13. Radical, 14. Postero-dorsal, 15. Antero-dorsal, 16. Laminal, 17. Apical, 18. Sub-apical

# Place of Articulation



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# Hindi Stops - PoA & MoA

|           | Voiceless<br>Unaspirated |                | Voiceless<br>Aspirated |                   | Voiced<br>Unaspirated |                  | Voiced<br>Aspirated |                     | Nasal |           |
|-----------|--------------------------|----------------|------------------------|-------------------|-----------------------|------------------|---------------------|---------------------|-------|-----------|
| Velar     | क                        | ka<br>/k/      | ख                      | kha<br>/kʰ/       | ग                     | ga<br>/g/        | घ                   | gha<br>/gʱ/         | ङ     | ṅa<br>/ŋ/ |
| Palatal   | च                        | ca<br>/c, t͡ʃ/ | छ                      | cha<br>/cʰ, t͡ʃʰ/ | ज                     | ja<br>/d͡ʒ, d͡ʒ/ | झ                   | jha<br>/d͡ʒʱ, d͡ʒʱ/ | ञ     | ña<br>/ɲ/ |
| Retroflex | ट                        | ṭa<br>/ʈ/      | ठ                      | ṭha<br>/ʈʰ/       | ड                     | ḍa<br>/ɖ/        | ढ                   | ḍha<br>/ɖʱ/         | ण     | ṇa<br>/ɳ/ |
| Dental    | त                        | ta<br>/t/      | थ                      | tha<br>/tʰ/       | द                     | da<br>/d̪/       | ध                   | dha<br>/d̪ʱ/        | न     | na<br>/n/ |
| Bilabial  | प                        | pa<br>/p/      | फ                      | pha<br>/pʰ/       | ब                     | ba<br>/b/        | भ                   | bha<br>/bʱ/         | म     | ma<br>/m/ |

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# Hindi Semivowels

य र ल व

Y

R

L

V

SEMI VOWELS

| Semivowels | Hindi | Punjabi | Bengali | Gujarati |
|------------|-------|---------|---------|----------|
| Y          | य     | ਯ       | য়      | ય        |
| R          | र     | ਰ       | র       | ર        |
| L          | ल     | ਲ       | ল       | લ        |
| V          | व     | ਵ       | —       | વ        |
| Z          | ज़    | ਜ       | য       | —        |

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# Hindi Fricatives

श

*śa*

ष

*ṣa*

स

*sa*

ह

*ha*

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# Hindi Half-Consonants

## Half consonants – Halant (हलन्त)

|    |    |    |    |    |
|----|----|----|----|----|
| क् | ख् | ग् | घ् | ङ् |
| च् | छ् | ज् | झ् | ञ् |
| ट् | ठ् | ड् | ढ् | ण् |
| त् | थ् | द् | ध् | न् |
| प् | फ् | ब् | भ् | म् |
| य् | र् | ल् | व् |    |
| श् | ष् | स् | ह् |    |

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# Hindi Vowels & Consonants

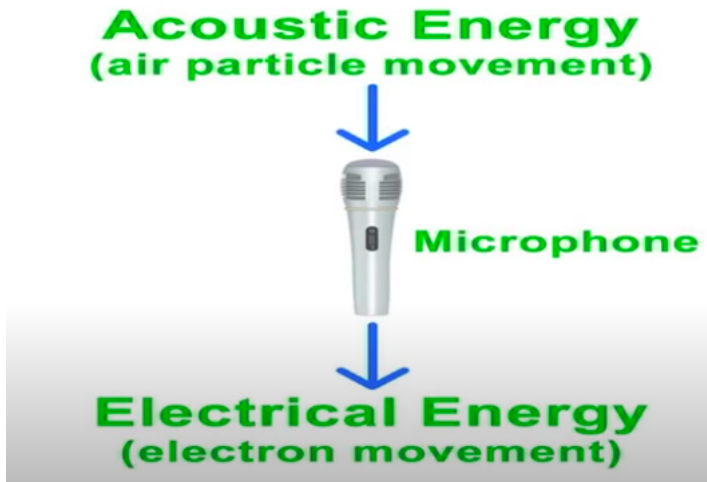
| Vowels   |     | Consonants |    |   |     |     |
|----------|-----|------------|----|---|-----|-----|
| अ        | a   | क          | ख  | ग | घ   | ङ   |
| आ ( ा )  | A   | k          | kh | g | gh  | n   |
| इ ( ि )  | i   | च          | छ  | ज | झ   | ञ   |
| ई ( ि )  | I   | c          | ch | j | jh  | n   |
| उ ( उ )  | u   | ट          | ठ  | ड | ढ   | ण   |
| ऊ ( ू )  | U   | T          | Th | D | Dh  | N   |
| ऋ ( ृ )  | R   | त          | थ  | द | ध   | न   |
| ॠ ( ॡ )  | RR  | t          | th | d | dh  | n   |
| लृ ( ॢ ) | lr. | प          | फ  | ब | भ   | म   |
| ए ( े )  | e   | p          | ph | b | bh  | m   |
| ऐ ( ै )  | ai  | य          | र  | ल | व   |     |
| ओ ( ो )  | o   | y          | r  | l | v/w |     |
| औ ( ौ )  | au  | श          | ष  | स | ह   | ज्ञ |
| अं ( ं ) | m.  | S          | sh | s | h   | jn  |
| अः ( ः ) | :   |            |    |   |     |     |

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# Digital Speech - Sampling Frequency, Bit Resolution

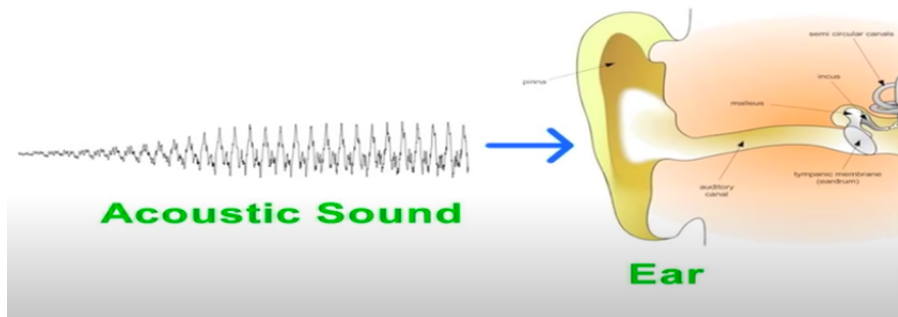
- Acoustic to analog electrical - transducer - microphone
- Electrical to acoustic - transducer - speaker
- Analog - discrete time signal
- Sampling process - sampling frequency
- Discrete - digital speech
- Analog to digital conversion - bit resolution

# Acoustic to Electrical Conversion



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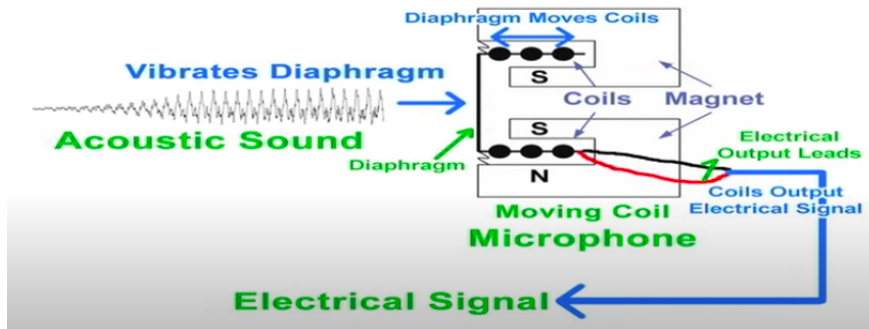
# Acoustic to Bio-Electrical Conversion



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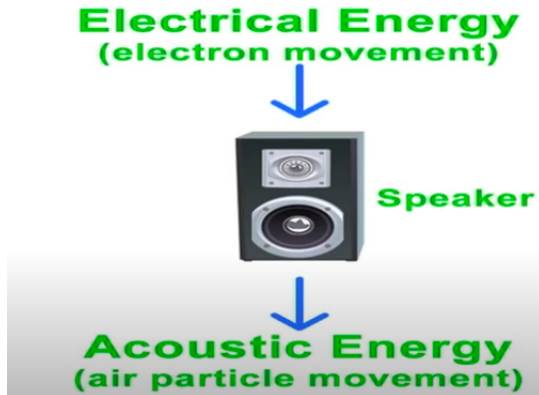


# Acoustic to Electrical Conversion



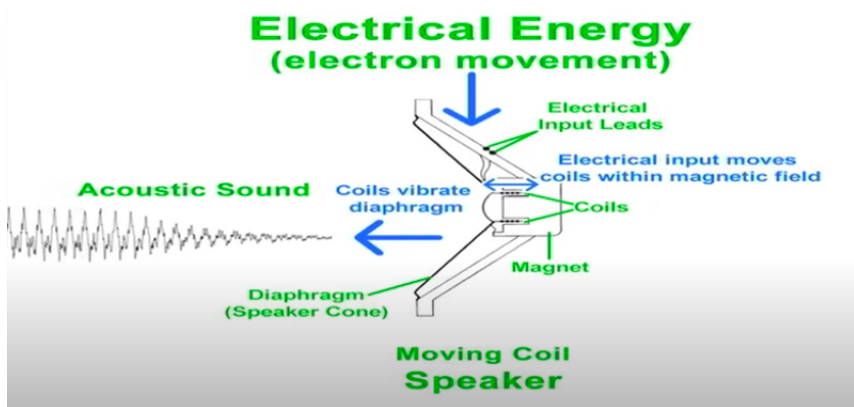
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# Electrical to Acoustic Conversion



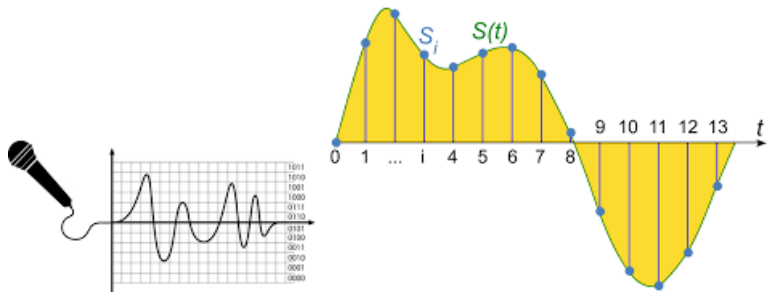
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# Electrical to Acoustic Conversion



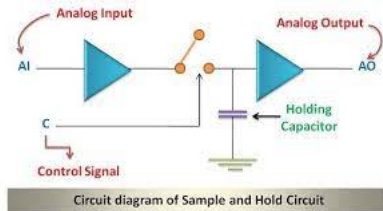
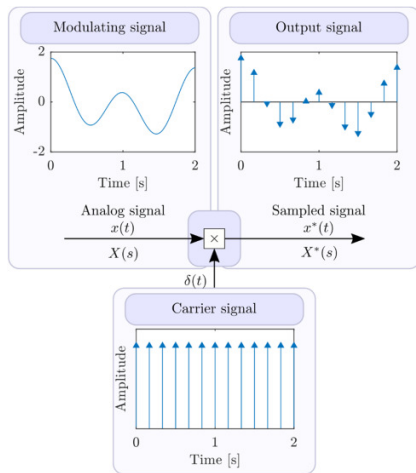
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# Analog to Discrete



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# Sampling Process

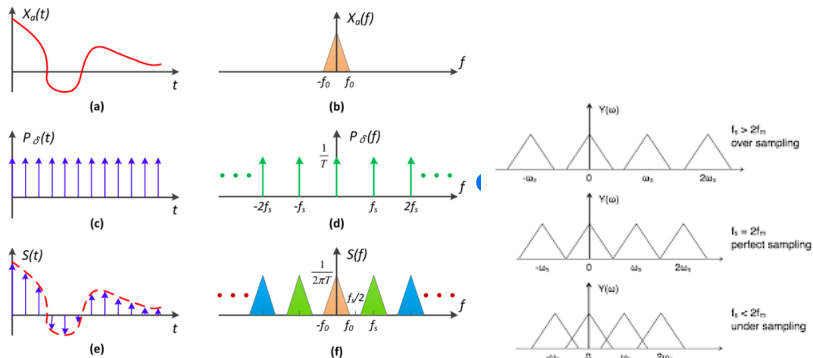


Electronics Coach

- Sampling Interval  $T_0$

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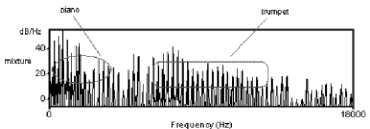
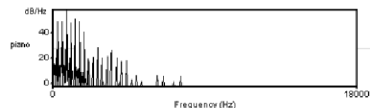
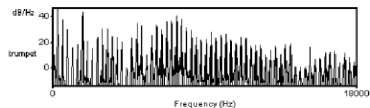
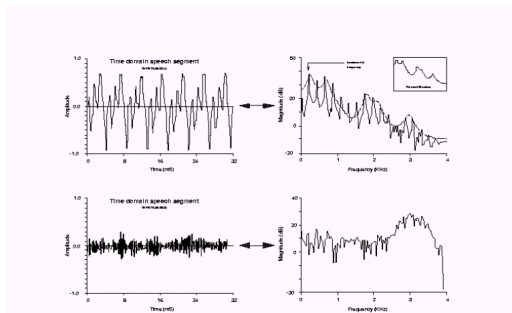
# Sampling Theorem



- Sampling frequency  $f_s \geq 2f_0$  or  $\omega_s \geq 2\omega_0$
- Sampling period  $T_0 = \frac{1}{f_s}$

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# Significance of Sampling Frequency in Speech Processing



3. Power spectrum of trumpet, piano and their mixture; frequency axis

- 8 kHz for vowel-like sounds, 16 kHz for fricative-like sounds and 44 kHz for music

# Discrete to Digital Conversion

- No. of Quantization levels  $q = 2^n$ ,  $n$  no. of bits in binary word (bit resolution)
- $n = 8 \implies q = 2^8 = 256$

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# Speech Signal Sampled at 44.1 kHz

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# Different Speech Segments Spectra Sampled at 44.1 kHz (1)

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# Different Speech Segments Spectra Sampled at 44.1 kHz(2)

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# Different Speech Segments Spectra Sampled at 8 kHz (Telephone / Mobile)

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# Speech Signal at 16 vs 8 bit resolution

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# Speech Spectra at 16 vs 8 bit resolution

Taken from public domain. Original authors are highly acknowledged.



# Speech Signal at 16 vs 4 bit resolution

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# Speech Spectra at 16 vs 4 bit resolution

Taken from public domain. Original authors are highly acknowledged.





# Speech Signal at 16 vs 1 bit resolution

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# Speech Spectra at 16 vs 1 bit resolution

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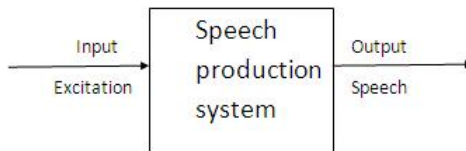
# Broad Segmentation of Speech

- **Three Regions** : Voiced, unvoiced and silence regions
- **Voiced** : Segments / regions of speech produced by voiced excitation (vibration of vocal folds)
- **Unvoiced** : Segments / regions of speech produced by unvoiced excitation (burst, frication, aspiration)
- **Silence** : Segments / regions of speech during which there is absence of any type of excitation

# Voiced-Unvoiced-Silence Speech Segments

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# Signal and System



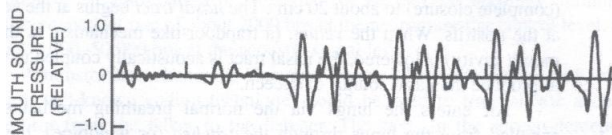
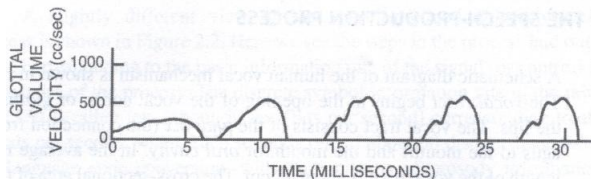
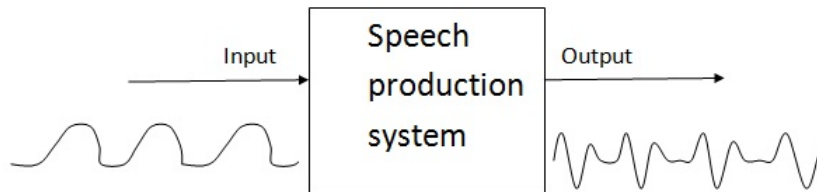
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# Human Speech Production System

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# Voiced Speech Production



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# Voiced Speech Segment

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# Unvoiced Speech Production

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# Unvoiced Speech Segment

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# Silence Regions in Speech

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# Silence Regions in Speech

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# Hindi Sound Units Classification

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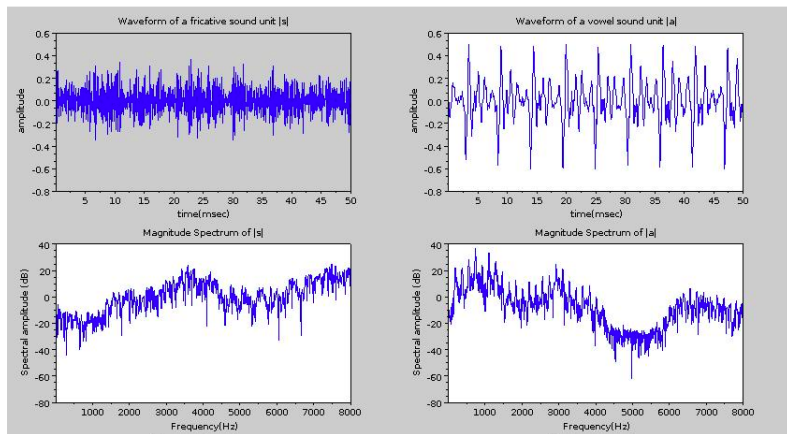


# Hindi Sound Units

| Vowels   |     | Consonants |    |   |     |     |
|----------|-----|------------|----|---|-----|-----|
| अ        | a   | क          | ख  | ग | घ   | ङ   |
| आ ( ा )  | A   | k          | kh | g | gh  | n   |
| इ ( ि )  | i   | च          | छ  | ज | झ   | ञ   |
| ई ( ि )  | I   | c          | ch | j | jh  | n   |
| उ ( उ )  | u   | ट          | ठ  | ड | ढ   | ण   |
| ऊ ( ू )  | U   | T          | Th | D | Dh  | N   |
| ऋ ( ृ )  | R   | त          | थ  | द | ध   | न   |
| ॠ ( ॡ )  | RR  | t          | th | d | dh  | n   |
| लृ ( ॢ ) | lr. | प          | फ  | ब | भ   | म   |
| ए ( े )  | e   | p          | ph | b | bh  | m   |
| ऐ ( ै )  | ai  | य          | र  | ल | व   |     |
| ओ ( ो )  | o   | y          | r  | l | v/w |     |
| औ ( ौ )  | au  | श          | ष  | स | ह   | ज्ञ |
| अं ( ं ) | m.  | S          | sh | s | h   | jn  |
| अः ( ः ) | :   |            |    |   |     |     |

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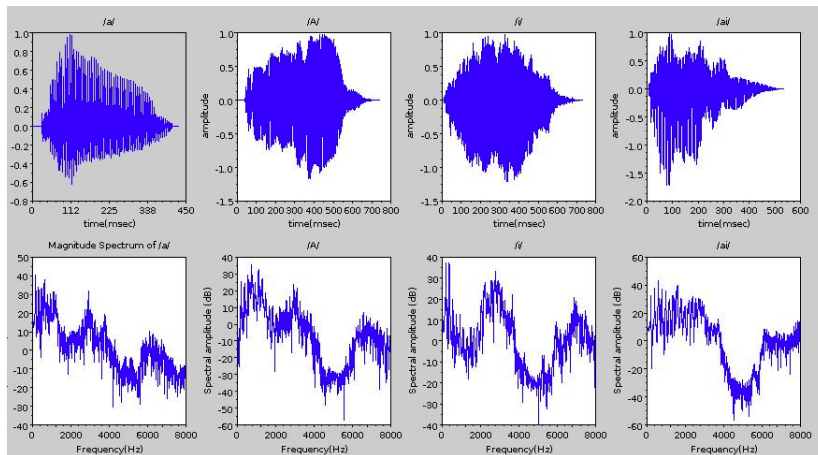
# Time & Freq Domain Repn of Vowels and Consonants



- Nearly voiced vs unvoiced excitation
- Low frequency vs Low to high frequency

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# Time & Freq Domain Repn of Short & Long Vowels

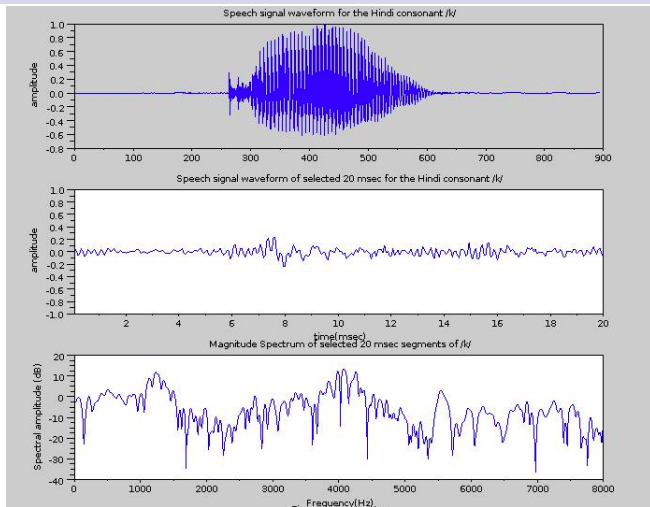


- Except time duration no difference between short & long versions

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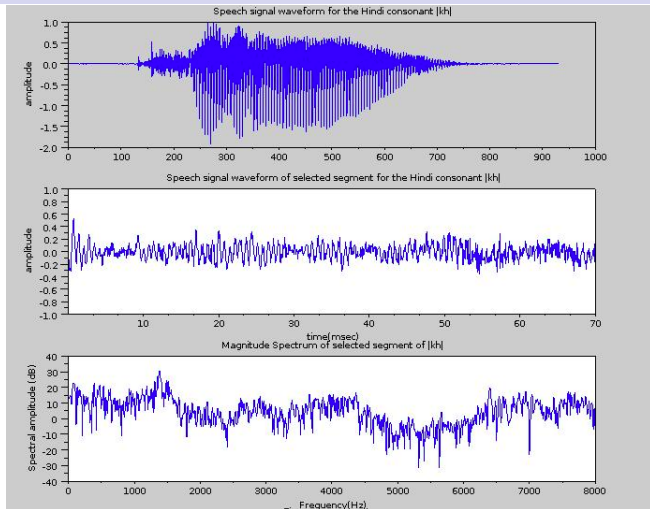
# Time & Freq Domain Repn of Velar Stop Consonant



- Stop consonant is low energy region prior to vowel

Taken from public domain. Original authors are highly acknowledged.

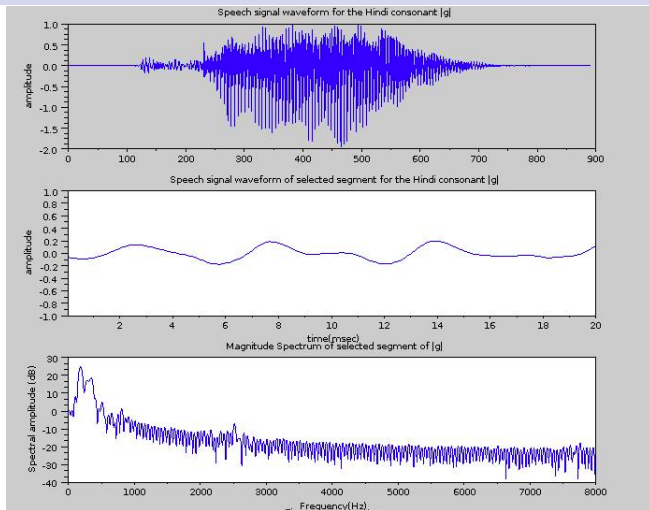
# Time & Freq Domain Repn of Velar Aspirated Consonant



- Low energy noisy region prior to vowel

Taken from public domain. Original authors are highly acknowledged.

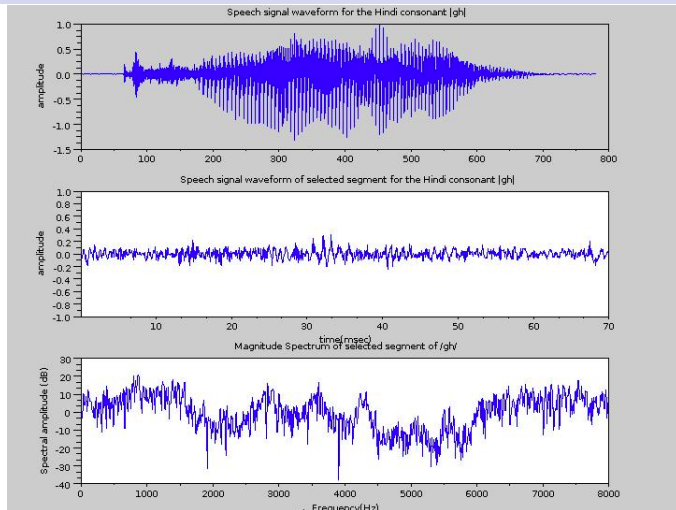
# Time & Freq Domain Repn of Velar Voiced Consonant



- Low energy nearly periodic region prior to vowel

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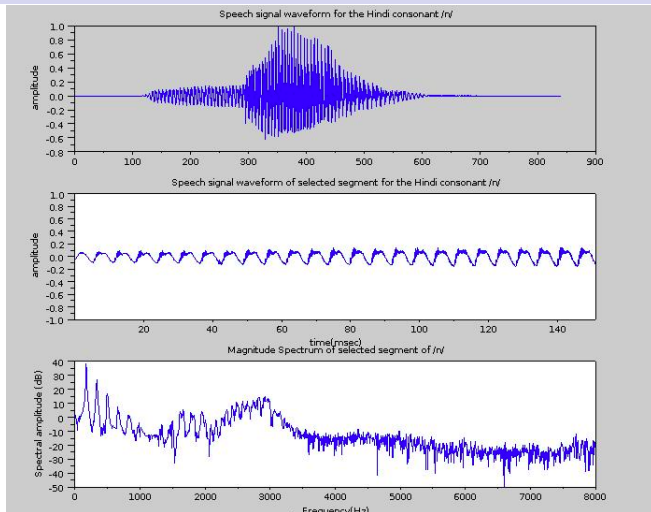
# Time & Freq Domain Repn of Velar Voiced Aspirated



- Noisy region with underlying voiced excitation prior to vowel

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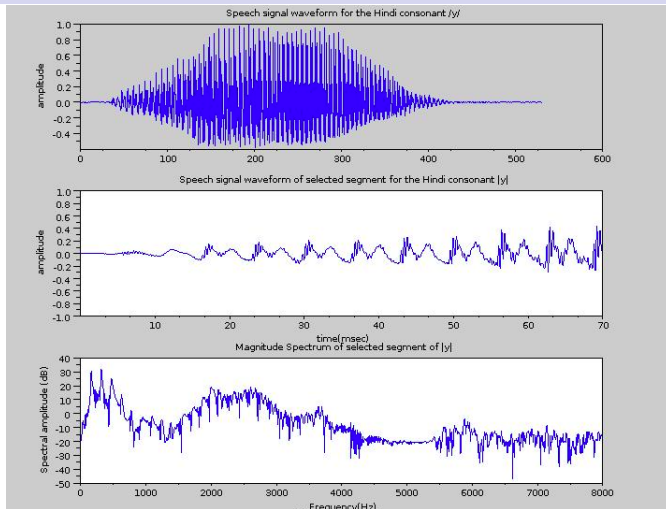
# Time & Freq Domain Repn of Nasal



- Relatively low energy voiced region prior to vowel

Taken from public domain. Original authors are highly acknowledged.

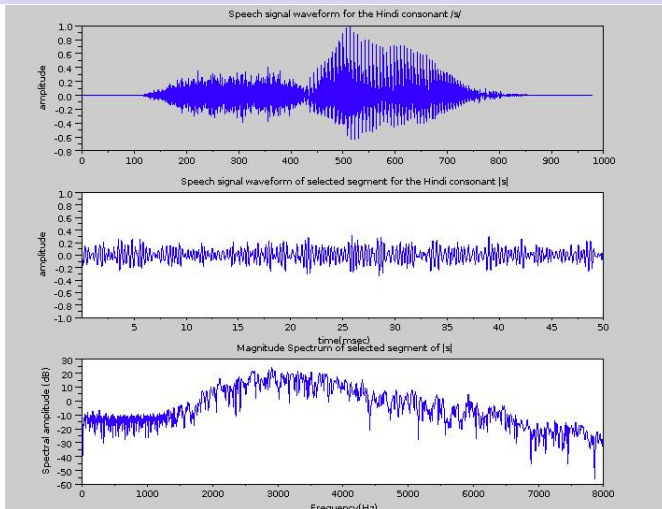
# Time & Freq Domain Repn of Semivowel



- Similar to vowel in TD & FD

Taken from public domain. Original authors are highly acknowledged.

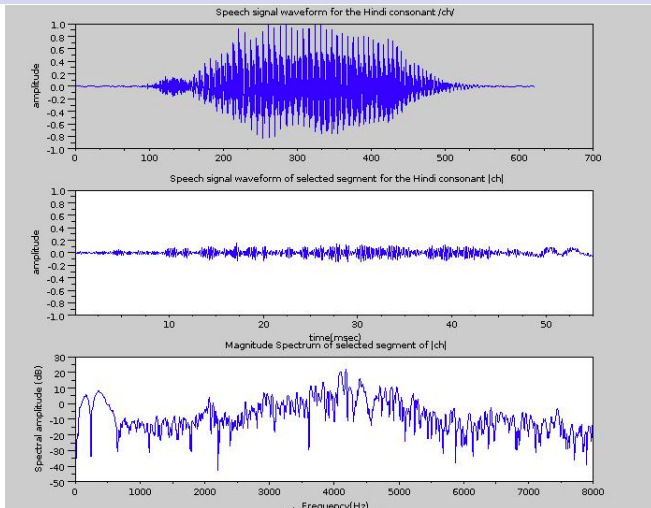
# Time & Freq Domain Repn of Fricative



- Noisy structure prior to vowel

Taken from public domain. Original authors are highly acknowledged.

# Time & Freq Domain Reprn of Affricate



- Shorter noisy structure compared to fricative prior to vowel

Taken from public domain. Original authors are highly acknowledged.



# Some Good Resources for LRS

- Chris Hart, “Doing a Literature Review - Releasing the research imagination” 2nd Edition, SAGE Publishers, Delhi 2018
- Arelene Fink, “Conducting Research Literature Reviews - From the internet to paper”, 5th Edition, SAGE Publishers, Delhi, 2020.
- Robert L. Jolles, “How to Run Seminars and Workshops”, Wiley, 2005
- Christine Harrington and Theresa Orosz, “Why the First-Year Seminar Matters: Helping Students Choose and Stay on a Career Path”, Rowman Littlefield Publishers, 2018.

# LRS Course Planning

- 2 credit course ( $1-0-0-4=2$ )
- 28 contact hours
- $2 \times 14 = 28$  hrs
- $2 \times 2 = 4$  hrs by instructor
- $2 \times 12 = 24$  hrs by student seminars
- Wednesday, 6.30-8.30 pm
- 2-3 slots for each student in a semester

# LRS Course Evaluation (1-0-0-4-2)

- Even though it is 2 credit course, going to be semester long course
- First instructional day ( $T_0$ )
- Except endsem, outstation candidates can take all other components of evaluation in online mode
- Midsem (15%) : As per calendar,  $T_0 + 7W$  ;
- Endsem (25%): As per calendar (Offline for all),  $T_0 + 14W$  ;
- Seminar Evaluation(40%): Continuous evaluation
- Literature Rev Report (20%): Continuous evaluation
- Attendance Policy : As per institute norms



# Seminars Evaluation

- 2-3 times seminar by each student
- Graded by course instructor, mentor/supervisor, committee of faculty members, entire batch of students
- Average score/grade will be announced after each seminar
- Final score / grade will be average of all seminars

- **Introduction:** Need for research, research vs course based study, qualifications for doing research, preparing for doing good research - course work, comprehensive, etc.
- **Literature Review:** Searching for articles, reading and comprehending articles, summarizing works reported in the articles, critical review of the existing works and identifying gaps.
- **Research Exploration:** Research questions, hypothesis for the novel direction, exploration for validating the hypothesis, results and discussion.

- **Patenting and Publications:** Importance of patenting and publications, steps for drafting patents, steps for writing scientific articles, significance of patenting and publishing in good venues.
- **Presentation, Report and Thesis Writing:** Preparing presentations, delivering seminars and writing reports, paraphrasing and giving credits/citations to the original author(s), thesis writing.
- **Conclusions and Future Scope:** Concluding current research work and identifying possible future directions.
- **Principles & Ethics in Research:** Significance of hard work, time management, health practices, patience, plagiarism and its consequences, and ethics in research.

# Introduction - Why Research?

- Stone age to generative AI
- Walking to supersonic jets
- Smokes, drum sounds and pigeons to 6G communication
- Isolated or villages in cylos to connected world (global village)
- Healthcare - No cure to advanced treatment
- Seed in someone's mind (research question), dream (hypothesis) and then to reality (research exploration)
- Social mindset - contributing something towards society vs Profit mindset - Invention and patents, royalty



# Introduction - Terminologies

- Research is systematic investigation into and study of materials and sources in order to establish facts and reach new conclusions
- Research is creative and systematic work undertaken to increase the stock of knowledge
- Research is to observe and understand what everyone else has done so far and do what none have done so far
- Research methodology vs research methods
- Original or fundamental or primary research
- Scientific research vs Applied or engineering research
- Invention vs innovation



# Introduction - Basic Realization

- Research degree or program is entirely different from course oriented program!
- Research exploration is different from laboratory courses
- Research is different from project course
- Research  $\implies$  Re-Search
  - Several persons like you earlier look at this question that you are thinking now to explore!
  - Did some work and documented, but unfinished agenda!
  - You are not the first & hence search again!
  - Why did not find soln like you? they did not look the way you did it
  - You will not be last & identify future directions



# Introduction - Preparing for doing research

- Appreciating the requirements of research based degree!
- Identifying broad field for research exploration
- Selecting a set of good courses and completing
- Identifying subfield for research exploration
- Literature review & critical review
- Listing different questions for exploration
- Hypothesizing different solutions possible
- Getting ready to jump-in

# Research based Degree vs Courses based Degree

- **Course based Degree:** A set of courses (subjects) organized in a pedagogical manner. Attend classes, submit assignments, write exams, carryout projects, get grades. Time bound program.
- **Research based Degree:** Set of good courses are only pre-requisites. Literature review, seminar, research exploration, patents or research articles, research presentations, report and thesis writing, defence. Not a time-bound program.
- Deterministic vs stochastic (probabilistic)
- One to many vs one to one
- Breadth vs depth

# Qualifications for Research based Degree

- Anybody with common sense can do research (ABCR)
- Need not be academic topper
- Willing to invest time and work hard
- Realizing research based degree is different to course based degree
- Having patience when you don't get results
- Walking slow is fine, but stopping completely is dangerous

# Selecting Good Courses and Investing Time

- For the broad research area, there are some courses whose deeper knowledge becomes crucial
- Discuss with supervisor, select set of courses
- Enroll to those out of this set and take course work seriously
- If courses are related to your research direction, will be useful for kick-starting some ground work related to research
- While doing course work, keep questioning / motivating yourself - courses knowledge will benefit your research
- Getting hold onto the core concepts taught in the courses is very important



# The Research Experience

- Way of life in research, relationship with your supervisor, and expectations and goals of program
- **What is involved in Research?** will have to solve some hard problem that nobody solved before!
- Maybe, in general research is about a lot more than problem solving.
- It is about conceptualizing, formulating research questions and directions to solve, exploration and critical insight.
- Problem solving is always there, but the role it plays varies.
- A common situation is to be developing some novel solution.
- Research questions, hypothesis and exploration.



# The Research Experience

- When you have novelty, publish
- After finding novel solution then wonder why nobody else did it.
- Reason is they did not ask that question, or look at it in that way.
- Don't look down on simplicity; good research is often simple.
- Your starting research project is not likely to be to solve some known hard problem, where one can make progress step by step.
- Don't start by thinking about a thesis or a thesis topic. Your goal is to produce papers initially.

# What can I get from Research?

- Research experience is about much more than learning to do deep work in some technical area.
- You should get a sense of confidence in the power of rational thought and the range of its applicability.
- After a research experience, you should have the inclination and ability to research anything, and the expectation that you will understand it.
- You should get the confidence and inclination to question all that is around you and seek out new ways of doing it or seeing it.
- You should be more likely to ask why things are done a certain way, and how it could be made better.



# What can I get from Research?

- Should give you confidence that you can jump into a new area, pick it up quickly, and have something interesting to say about it.
- More than depth in any one area it should give you the courage to jump from area to area.
- You might increase your appreciation for creativity, in other people and in all areas of life.
- You might view or think differently about anything you encounter, appreciative of how it departed from the previous works.
- You should learn to value creativity and seek it out.
- You should be constructively critical about how things are currently and how we can improve.



# The Path to Research: Supervisor & Scholar

- One to one interaction vs classroom
- Each supervisor style and scholar preparedness are different
- It is your research problem vs do what I say!
- Wide range and how to cope up?
- End goal is to make you independent researcher
- **Phase I:** Well defined problem by supervisor, work judiciously to make a good publication
- **Phase II:** Fuzzy problem by supervisor, explore yourself to make a good publication
- **Phase III:** You are on your own



# The Path to Research: Meeting with Supervisor

- At least once a week
- Any urgent discussion, always welcome by taking appointment
- Working together is fun.
- Both supervisor and scholar should enjoy.
- If you want to discuss a problem don't keep it to yourself.
- A good deal of research is spontaneous and social, arising from interactions with your peers or advisor.
- You may not have something substantial to report at a meeting. That's OK, as long as it's not a habit.

# The Path to Research: Meeting Preparation

- Prepare for your meetings.
- Prepare the presentation so that you can make it clearly and well.
- Plan the order in which you will say things.
- Communication goes better and quicker if you prepare well, and this makes better use of our limited time.
- If fuzzy and need more time for understanding, you get lower quality feedback and it is harder to work together towards a solution.
- Brief about previous meeting, progress made and seek suggestions.
- Think of your presentations as practice for larger audiences.
- Preparation for presentation diff. from actual technical work.
- Not all meetings need presentation, don't go overboard either.



# The Path to Research: Meeting Notes

- Go with a notebook to meetings and take notes.
- Don't hesitate to spend time taking note.
- Ask for clarifications or repetitions if necessary.
- Make sure clear with each other about the technical issues and what to do next.
- Try to be clear about where you think you are headed in the period up to the next meeting.
- What are the tasks, questions, deliverables, if any? Have this written down if you are unsure you will remember it.



# Learning to do Research

- **Learning by thinking:** The first rule of research is to think, think and think again. Thinking is fun. If you don't find it so, it's an indication you are in the wrong business.
- **Learning by example:** You pick up how research is done by seeing examples and extrapolating.
- **Natural learning:** Who are positive towards learning. Appreciate existing things and positively critical for improving.
- **Understanding versus knowledge:** It is more important to understand well what you know than to know a lot. Successful research comes from having a good understanding and converting to knowledge.

# Gauging Progress: Am I on track?

- Understanding, knowledge creation, publications
- Research work is not job
- Phase by phase improvement is convinced internally
- Quality publications are important
- **No. of publications:** At least 2 Q1 Journals for PhD, 1 Q1 Journal for MTech(R) and 1 Q1 for UGR
- Develop communication skills, both written and verbal. Able to plan and give clear talks and to write clear and correct technical papers.
- **Time Duration:** 4-5 years for PhD, 2-2.5 years for MTech(R), 1 year for UGR.



# Time Allocation

- How people allocate research time varies
- Some work on an inspiration or deadline-driven basis, sometimes putting in long hours, then doing nothing for a few days.
- Others maintain a steady schedule, coming to the lab at a certain time in the morning and leaving a certain time in the evening.
- Find what works best for you.
- Long-term productivity is important - like a summer, a quarter, or more typically, a year.
- If a long period passes without significant progress, you should worry.





# Working with others

- Good research needs a collaboration.
- You learn better and produce more by interacting with others.
- Besides of course talking to your advisor, talk to other students and engage in joint work with them.
- Of course you must find a good balance between your own pace and that of others.

# Is research degree for me?

- Is going for a research based degree the right thing for me?
- If so, is the area I am pursuing now right for me?
- If so, is the supervisor I have now right for me?
- Don't kid yourself
- Am I smart enough?
- Self confidence
- Choosing area and supervisor

# Comprehensive Examination

- What is comprehensive examination?
- Why it is needed?
- **Comprehensive** : Complete and including everything necessary.
- **Comprehensive Examination**: Check the preparedness of candidate(s) to carryout good research.
- **Why it is needed?**: Ex. Engineering research. To check clarity on fundamentals of the field of research, required mathematics, etc.
- Better preparedness will give deeper understanding and generalization of research findings during research exploration



# Comprehensive Examination

- Unlike other exams, comprehensive exams are not linked to any particular course, but rather test knowledge across one or more general fields of study.
- Most commonly come after student has completed required coursework and before starting on research work.
- Successful passage is sometimes required for a student to be considered a Ph.D. candidate
- Purpose of exam is to ensure student is familiar enough with her area of research to make original contributions.
- No standard format. No exam to quite rigorous.
- Informal meeting of few hours; critical review of one's academic portfolio; submission of an academic paper; series of proctored exams followed by oral exam



# Summary of Introduction to Research

- Course planning
- Syllabus
- Why research?
- Preparing for doing research
- Course vs research based degree
- Good courses
- Path to research
- Supervisor and scholar
- Time allocation
- Assessing progress



*Thank You*